

Homework 3

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##Set Up

```
# reading in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
```

```
# creating object for kelp data
kelp <- read_csv("temp-kelp.csv")
```

Rows: 32 Columns: 2

-- Column specification -----

Delimiter: ","

dbl (2): temp_c, kelp_elong

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# creating object for personal data
commute <- read_csv("envs193ds-personal_data - Sheet1.csv")
```

Rows: 21 Columns: 6

-- Column specification -----

Delimiter: ","

chr (4): day of the week, weather, construction (y/n), date

dbl (2): time of day (hr:min), commute length (min)

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Problem 1

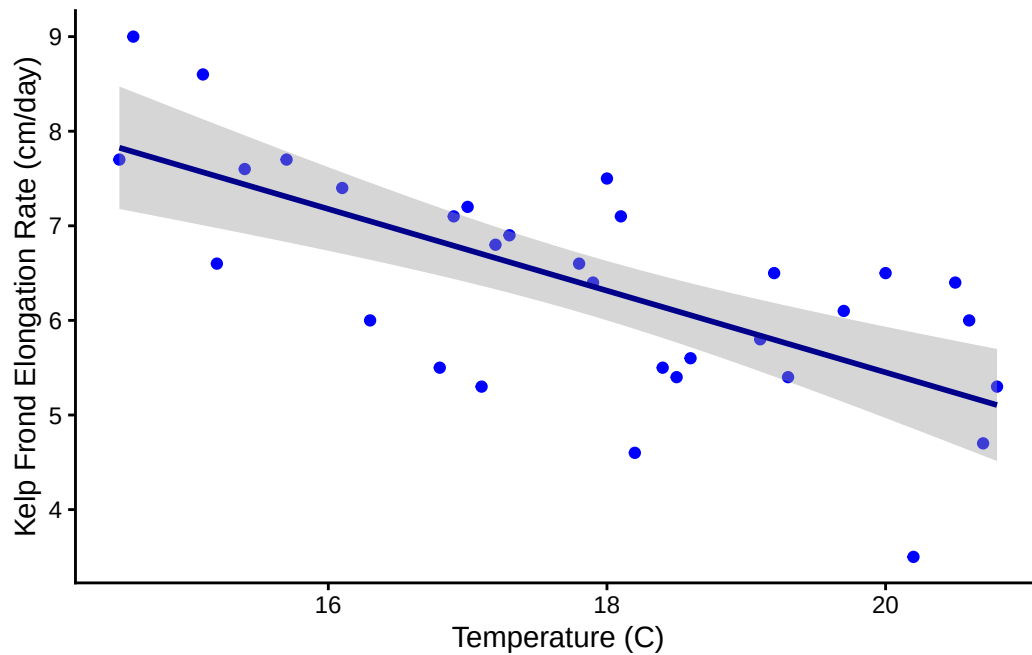
a.

The two tests that we should run to determine the relationship between temp. and giant kelp elongation rate would be a Pearson correlation test and a Spearman rank test. The tests both measure the strength of the relationship between the variables but the Pearson test relies on a normal and linear distribution and the Spearman test does not need normal distribution.

b.

```
# base layer ggplot
ggplot(kelp, mapping = aes(x = temp_c, y = kelp_elong)) +
  # first layer: scatter plot
  geom_point(color = "blue") +
  # second layer: line of best fit + ribbon
  geom_smooth(method = "lm", se = TRUE, color = "darkblue") +
  # renaming x and y axis
  labs(x = "Temperature (C)", y = "Kelp Frond Elongation Rate (cm/day)") +
  # changing the theme of the graph to classic
  theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



c.

Checking assumptions

Linear = yes the data does have a linear relationship. You can see this in problem 1b where the linear graph is the right model for the data.

Normality = Yes the data is normally distributed, this is shown through the Shapiro test.

Based on these two assumption checks the correct test to run would be the Pearson correlation test.

```
shapiro.test(kelp$temp_c) # running normality test for temp
```

Shapiro-Wilk normality test

```
data: kelp$temp_c
W = 0.96254, p-value = 0.3219
```

```
shapiro.test(kelp$kelp_elong) # running normality test for kelp growth rate
```

Shapiro-Wilk normality test

```
data: kelp$kelp_elong  
W = 0.98756, p-value = 0.966
```

running the test

```
# running the correlation test  
cor.test(kelp$temp_c, # setting first variable  
         kelp$kelp_elong, # setting 2nd variable  
         method = "pearson") # setting the method to be Pearson
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong  
t = -5.189, df = 30, p-value = 1.366e-05  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
 -0.8359665 -0.4460157  
sample estimates:  
      cor  
-0.6877489
```

d.

In order to test the strength of the relationship between temperature in Celcius and kelp frond elongation rate in cm/yr I used a Pearson test of correlation ($r(30) = -0.687$, $p = 1.36 \times 10^{-5}$, $\alpha = 0.05$, $CI = (-0.85, -0.45)$). This test was ran because the data was normally distributed and showed a linear relationship. This test yielded a value of $r = -0.69$, which shows a moderate, negative linear correlation between variables. Because of this value and the p value being less than 0.05 we can safely reject the null hypothesis on the grounds of there being a negative correlation between the variables.

e.

The results seem to show that the increasing temperature of the ocean may cause kelp growth rates to slow down, which would in turn have larger effects on the kelp's overall health. When the kelp was grown in warmer temperatures the kelp grew at a slower rate. The relationship

is represented by a value of -0.69 which means that there is a moderately strong, negative correlation between temperature and elongation rate.

f.

```
# setting up the correlation test
cor.test(kelp$temp_c, # setting first variable
         kelp$kelp_elong, # setting 2nd variable
         method = "spearman") # assigning spearman test
```

Spearman's rank correlation rho

```
data: kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

Through running a Spearman rank test ($\rho = -0.69$, $p = 1.29 \times 10^{-5}$, $\alpha = 0.05$) I determined that there is a strong negative relationship between temperature and frond elongation rates. The two tests have both led me to reject the null hypothesis of there being no correlation between variables because they both show a negative relationship between the increasing temperature and decreased growth rates. The Spearman test tested without normality and found there is a negative relationship, and the Pearson test used normality and also found a negative, linear relationship.

Problem 2.